

A Two-Stage Stochastic Optimization Model for Pediatric Vaccine Tender Scheduling

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1 *Dataset and scenario generation*

NU: The model incorporates synthetically generated data and based on real world historic data. We focus our analysis on the antigens in Table 1. We use the Linksbridge (Linksbridge) Vaccine Almanac datasets, which provides information on vaccine demand in LIC and LMIC in the next 10 years.

Vaccine Name	Antigens
MMR	Measles, Mumps, Rubella
Penta, Hexa	Diphtheria, Tetanus, Pertussis, Haemophilus influenzae type b, Polio, Hepatitis B
Rotavirus	Rotavirus
HPV	Human papillomavirus
PCV	Pneumococcal conjugate

Table 1: Vaccines and corresponding antigens

We used historical data related to the cost of vaccines, which UNICEF (UNICEF) publishes. Since the cost of a vaccine dose differs by manufacturer, we use the corresponding average cost per dose in the model. The reservation price of a vaccine represents the price at which UNICEF purchases vaccines from manufacturers in a specific market. We assume that

the data presented by UNICEF, the published price per dose, represents this reservation price.

1.0.1 Stochastic Demand

In the linear program, we utilize data forecasts by vaccine from UNICEF (UNICEF). As shown in Figure 1, we see an example of vaccine demand over a ten-year period. These forecasts, however, do not approximate a distribution and serve only as point estimates. We, therefore, generate forecast data to represent the stochastic nature of the input, as described in Mitra (2006).

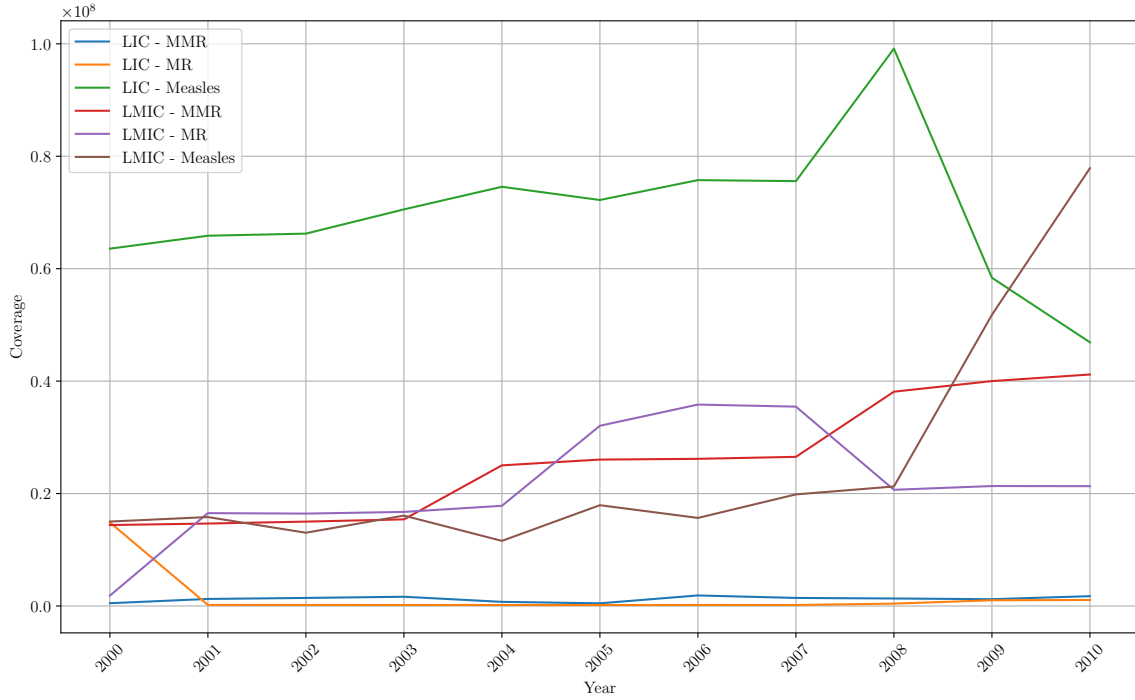


Figure 1: Measles Containing Vaccine Demand

Vaccine tenders are generated for antigens. It is, therefore, necessary to translate vaccine demand into antigen demand. To generate antigen demand by year, we aggregate demand for each antigen, respective to the total sum of each vaccine $\mathcal{V}_a$ in the vaccine set $\mathcal{V}$. Measles demand, for instance, is the sum of all vaccines that contain the measles antigen (Measles, MR, MMR). From this aggregation, we utilize standard time series forecasting methods. Data for each antigen does not present ideal forecasting data. The data

is therefore transformed to remove elements of seasonality, ensure the data is stationary (remove trends), and finally ensure normally distributed residuals. We fit varying ARIMA and linear models, as described in Bowerman et al. (2005), to the data, described in Table 2. ARIMA proved to be an effective time series analysis method to forecast our data with trends, while linear trend growth is modeled utilizing standard linear modeling. In this study, we utilized the `pmdarima` package in Python to generate such forecasts (Taylor et al. (2019)). Transformations for ARIMA forecasts are selected, such that the transformed data generates geneerally normally distributed studentized residuals.

Antigen	Transform	ARIMA Model
Measles	SQRT	(2,2,1)
Mumps	SQRT	(0,2,1)
Rubella	SQRT	(2,2,1)
Diphtheria	Log	(0,2,1)
Tetanus	Log	(0,2,1)
Pertussis	Log	(0,2,1)
Hib	None	(0,2,1)
Hep B	SQRT	(1,0,2)
Polio	Log10	(0,2,1)
PCV	$\wedge 0.75$	(0,2,0)
Rota	SQRT	(1,2,0)
HPV	None	(2,2,0)

Table 2: Antigen Demand Forecast Models

Models selected further ensure that the forecast data do not retain seasonality or trend. From these models, we determine a 95% confidence interval for the data. Using the upper and lower bounds and their standard deviation, we generate random normal data, bounded at each year by the confidence interval. An example forecast is shown in Figure 3. As the time periods of the forecast progress, we note an increase interval, due to increasing standard deviations.

500 initial demand scenarios are generated and then reduced to a weighted set. From this set, the associated weights correspond to the probability of scenario occurrence, such

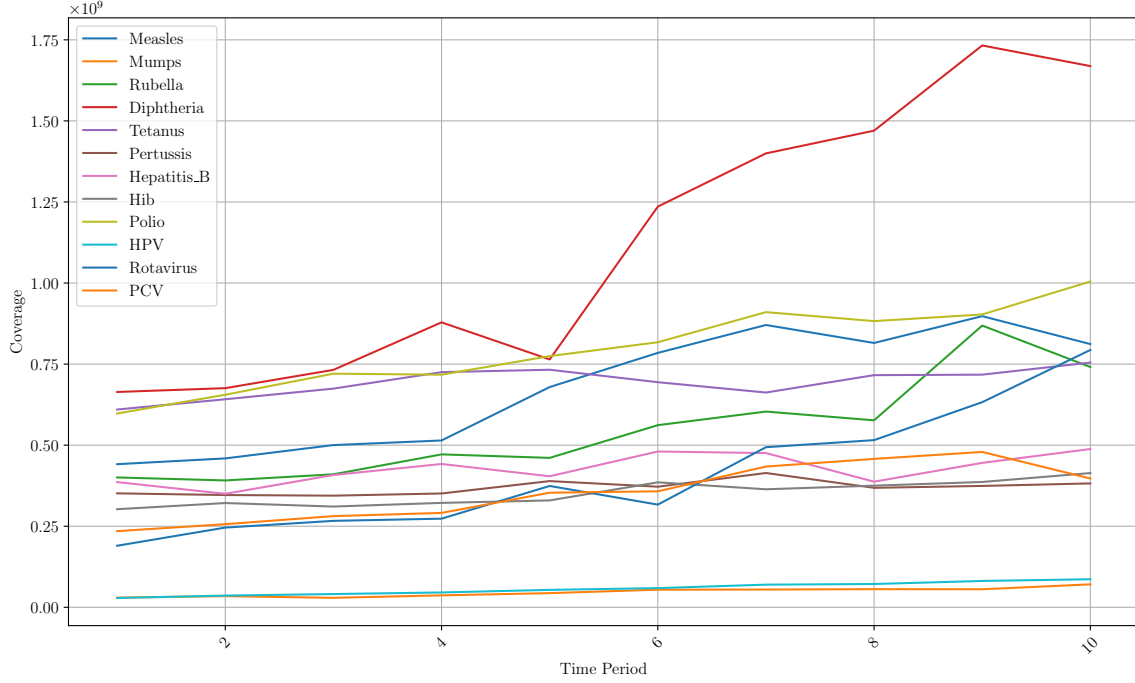


Figure 2: Vaccine Demand Forecast

that the set represents a total sum of 1. Forecast examples are shown on the left of Figure 2, while the reduced weighted set is shown on the right.

1.0.2 Stochastic Production Capacity

Production capacities, which often are confidential information, are synthetically estimated. In this analysis, we assume that capacities reflect previously delivered vaccine amounts, per producer. The production capacity baseline is estimated as the total capacity for a given producer in 2024, based on the Market information for access to vaccine (M4IA) database World Health Organization (a). From this database, we calculate the approximate percent market share for each manufacturer based on the average volume of vaccines previously supplied, and scale the capacity relative to the antigens included in this study. The scaled production capacity estimate represents the baseline production capacity for this analysis.

From the World Health Organization (b) WHO Vaccine supply and logistics data repository, the vaccine stock-out dataset was analyzed. Data for the years 2003 to 2023 was analyzed. Though events were reported between 2003 and 2015, their specific cause was

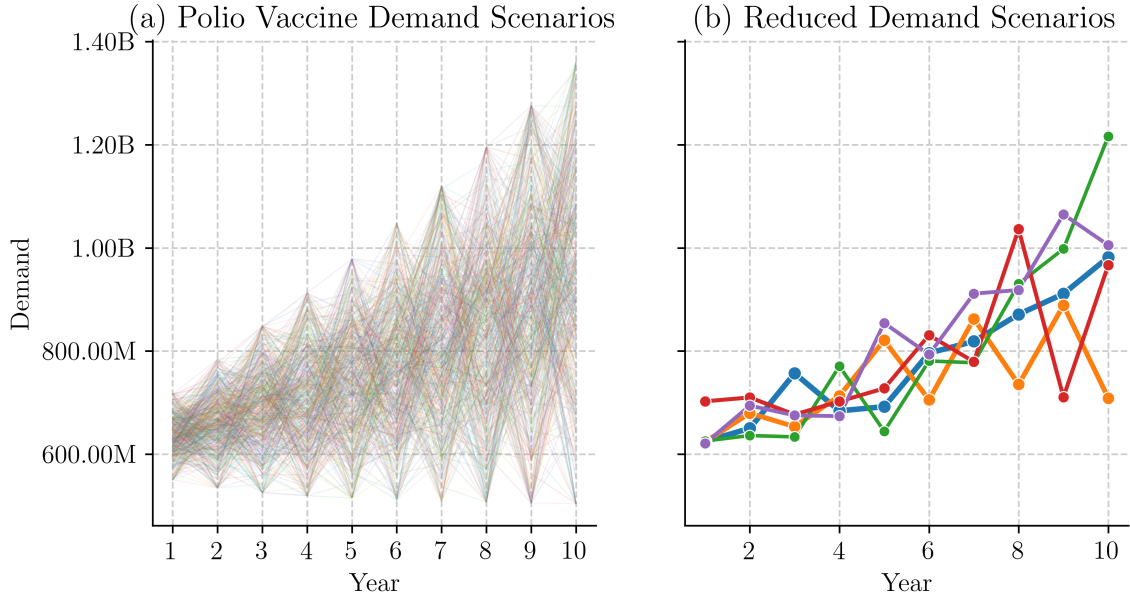


Figure 3: Vaccine Demand Forecast

not annotated in the data. Data from 2016-2024 contains stock-out descriptions for (PERCENT) of the data. Data from this period containing ND (no data), NR (not recorded), or blanks was omitted for clarity. Table 3 describes those events and their associated probability of occurrence from the data.

Scenario	Probability	Effect
None	84.18%	
Supply Chain Issues	0.58%	5%
Funding delays	4.12%	8%
Vaccine shortage	1.73%	10%
Inaccurate forecasts	0.23%	5%
Other/Not identified/Not known	2.75%	7%
Procurement delays	3.66%	8%
Pandemic	2.75%	44-32-8%

Table 3: Frequency of Event Occurrence

Given events and their probability of occurrence, the estimated production capacity baseline is modified to reflect potential supply disruptions. Scenario data baseline capacity

in year 1 (2024) is assumed to be constant throughout model years, and is only affected by events in scenarios. Increases in production capacity are induced within the model, through the corresponding constraints and coefficient in the objective function.

1.0.3 Scenario Pair Generation

Given n antigen demand scenarios, and m production capacity scenarios, $n \times m$ scenario pairs are generated from the data above. The probabilities associated with both antigen demand, and production capacity scenarios are combined, to generate overall scenario pair probabilities, for each $n \times m$ scenario pair in the set ω , whose total probability equals 1.

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